

Mapping the First Hop: Denominators, Hierarchy, and Gateway Structure in Reported Indian Foreign Ownership Networks

Canonical research synthesis

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Abstract

This paper reconstructs the reported foreign ownership networks of 28 Indian parent buckets in a supplied research dataset. The source is not a firm cross-section: 3,742 candidate source rows reduce to 3,567 preferred target-years, 1,834 structural target occurrences, 1,830 parent-scoped normalized entity candidates, and 1,818 global normalized candidates. Keeping these denominators separate changes both rankings and interpretation. The modal registration or mapping channel contains 59.7% of targets in the pooled data and 62.1% under equal-parent weighting. Among 24 sufficiently large parent groups, the median largest observed subtree contains 43.2% of normalized entities; the corresponding strict directed-acyclic-graph dominator estimate is 41.7%. First-hop gateways in the Netherlands have a mean of 34.8 downstream entities, compared with a parent-equal mean of 37.1, and a leave-one-parent-out range of 21.2–39.7. Direct relations are also commonly international: 951/1,650 nonroot edges (57.6%) cross a jurisdiction boundary, with an equal-parent estimate of 58.7%. These findings survive entity deduplication, parent balancing, leave-one-parent-out checks, and alternative network definitions. Other apparent patterns do not. Reported level and reconstructed graph distance disagree for 236/1,687 complete paths, and pooled depth, ownership, sector, and financial patterns are strongly affected by parent composition or measurement limitations. The paper therefore establishes a descriptive topology for this dataset, not causal, legal, tax, regulatory, or misconduct conclusions.

Keywords: ownership networks; corporate hierarchy; subsidiary geography; entity resolution; robustness; denominator audit

Scope note. This first canonical version deliberately uses only evidence validated inside the supplied data and code. It does not add an external literature review or institutional interpretation, because those sources were not part of the adjudicated evidence base. Such context is a task for a separately sourced revision.

Contents

1	Introduction	3
2	Data and analytical units	3
2.1	Source structure and scope	3
2.2	Targets, normalized entities, and mapping channels	4
2.3	Financial observations	4

3	Ownership-network construction and estimands	4
3.1	Edge resolution	4
3.2	Paths and depth	5
3.3	Network concentration	5
3.4	Geographic roles	6
3.5	Weighting and robustness	6
4	Denominators and parent architecture	7
4.1	Raw-row weighting changes the sample composition	7
4.2	Architecture is multidimensional	7
5	Dominant channels and large subtrees	9
6	Geography and intermediary structure	11
6.1	Cross-border ownership relations	11
6.2	First-hop gateway amplification	11
6.3	Residence, intermediary prevalence, and path exposure	11
6.4	Why a fixed “centre” category is not a core result	12
7	Reported depth and observed graph distance	15
8	Ownership fields, financial coverage, and structural proxies	16
8.1	Recorded stakes and the ownership-depth gradient	16
8.2	Financial attrition and extraction reuse	16
8.3	A limited name-based structural proxy	17
9	Robustness, heterogeneity, and evidence classification	18
10	Limitations and interpretation boundary	20
11	Conclusion	20
A	Exact estimands and counting rules	22
A.1	Units	22
A.2	Missing values and duplicates	22
A.3	Subtree and path denominators	22
B	Depth audit	23
C	Ownership-field audit	23
D	Evidence disposition	23
E	Reproducibility	25

1 Introduction

Datasets on international corporate structure often combine objects that look deceptively interchangeable. These include source records, reporting years, registered investments, named subsidiaries, parent relations, and complete ownership paths. Treating any one of these as “the number of firms” can mechanically change which parent groups or jurisdictions dominate a result. Hierarchy adds a second problem. A location can be common because many entities reside there, because a smaller number of upstream entities have many descendants, or because one unusually large parent group repeats the same route. Those are different empirical statements.

This paper asks a deliberately bounded question: how do the parent groups in the supplied 28-parent dataset organize their reported foreign ownership networks across mapping channels, hierarchy, and jurisdictions, and which patterns remain after correcting units, duplicates, parent weights, missing nodes, and graph definitions? The goal is measurement-corrected description. The data do not identify why a structure exists, whether it is current, whether a named entity has a particular legal function, or whether any route has a tax or regulatory purpose.

The analysis makes five contributions within that boundary. First, it builds an explicit denominator ladder from source observations to target, entity, edge, path, mapping-channel, and parent units. Second, it reconstructs a parent-scoped directed ownership graph while retaining named-but-unscraped parents and their supplied countries. Third, it separates resident geography from intermediary and descendant exposure. Fourth, it reports pooled, equal-parent, leave-one-parent-out, and alternative-definition estimates side by side. Fifth, it treats failed hypotheses as results: several initially striking depth, ownership, sector, and financial relationships do not support a general interpretation after the relevant correction.

The central evidence is as follows. Parent networks are heterogeneous along dimensions that cannot be collapsed into a single defensible complexity ranking. Many are concentrated behind one mapping channel or one large observed subtree. Dutch and Mauritian first-hop entities have greater downstream amplification than their resident counts alone would suggest, while the United States is primarily prominent as a resident location. A majority of reported nonroot edges cross jurisdictions, and that result is unusually stable to parent balance. In contrast, reported deep hierarchy is concentrated in a few parent groups and differs materially from reconstructable graph distance. The financial layer is sparse and extraction-contaminated, so it supports an audit of measurement quality rather than comparative financial performance.

The remainder of the paper defines the units and graph, documents the denominator correction, presents parent and jurisdictional topology, examines depth and financial limitations, and concludes with an explicit evidence boundary.

2 Data and analytical units

2.1 Source structure and scope

The immutable input is a Stata file containing candidate financial-statement sources attached to reported foreign targets. The file has 3,742 rows and 122 standardized jurisdiction labels across 28 supplied ultimate-parent buckets. Rows are not unique companies. A target can recur for multiple fiscal years and can have multiple candidate sources within a year. The upstream preferred-source flag selects 3,567 target-year rows, but structural fields are invariant within each target identifier. The structural cross-section therefore contains 1,834 target occurrences.

The sample is not accompanied by a sampling frame for all Indian outward investment. No result below is generalized beyond the supplied parent buckets. Similarly, the file provides reported names

Table 1: The canonical denominator ladder

Analytical unit	Count	Definition
Raw source rows	3,742	source candidate
Preferred target-year rows	3,567	$\text{target_id} \times \text{fiscal_year}$
Structural target occurrences	1,834	target_id
Parent-scoped normalized entity candidates	1,830	$\text{parent} \times \text{normalized name-country}$
Global normalized entity candidates	1,818	$\text{normalized name-country}$
UINs	186	uin
Observed level-0 targets	184	target occurrence
Ultimate-parent buckets	28	source parent label
Complete observed paths	1,687	target path

Notes: Counts are generated from the immutable source by the canonical pipeline. A target occurrence is one reported ownership-path target. Normalized entities are conservative name-country candidates, not verified legal identities. A complete observed path reaches the Indian parent through observed entity nodes.

and ownership relationships, not a complete legal-entity registry or verified historical ownership charts.

2.2 Targets, normalized entities, and mapping channels

The primary path and exposure unit is the target identifier. For within-parent architecture, entity names are normalized conservatively: Unicode is reduced to ASCII, case and punctuation are removed, ampersands are standardized, and the supplied country is spelling-harmonized. A parent-scoped entity key is the combination of ultimate-parent bucket, normalized name, and standardized country. This yields 1,830 candidates, only slightly fewer than the 1,834 targets. Dropping the parent component yields 1,818 global candidates. The difference is small in aggregate, but the estimands are conceptually distinct: a shared venture may be one global entity while still representing exposure in more than one parent network.

The file also contains 186 UIN values. They are retained as supplied and treated only as registration or mapping channels. The analysis does not infer the semantics of their substrings, a registration date, or legal-entity identity. 184 UINs have an observed level-0 target, so two mapping channels lack an observed first-hop entity in this file.

2.3 Financial observations

Financial variables remain at the preferred target-year level. Coverage gates are reported separately from network construction. A target has a located source if any source row carries the supplied source-found indicator, a parsed balance sheet if equity is numeric in any row, and a valuation-ready record if the upstream ready flag is present at least once. Absolute monetary values are not pooled across currencies or unspecified units.

3 Ownership-network construction and estimands

3.1 Edge resolution

Each structural target supplies an entity name and country, reported level, immediate-parent name and country, UIN, and ultimate-parent bucket. Candidate parents are formed within the same

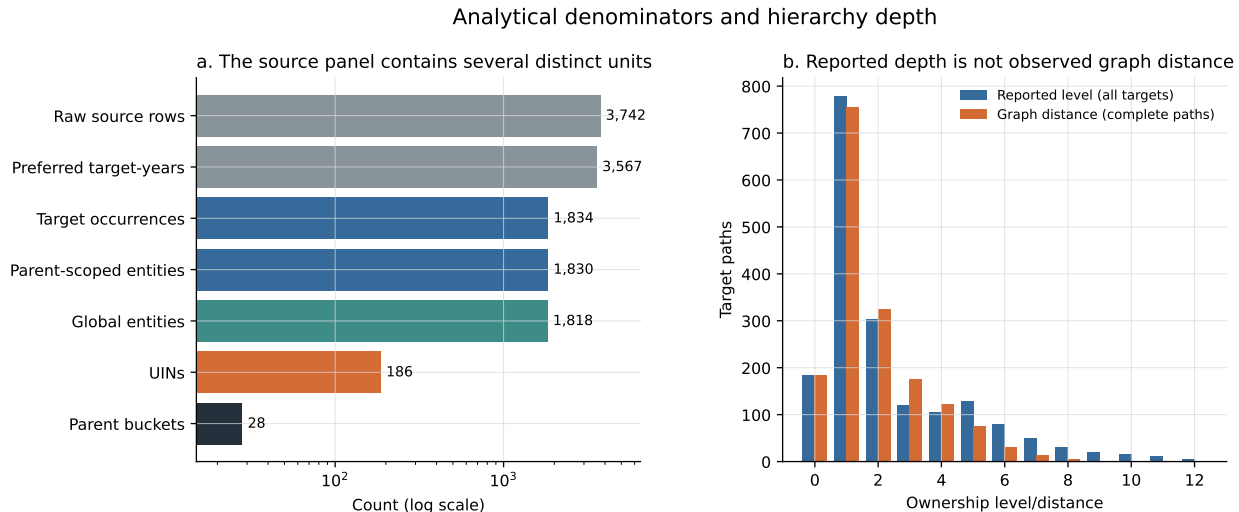


Figure 1: Analytical denominators and two measures of hierarchy depth

Notes: Panel (a) uses a logarithmic count scale because the units differ sharply in size. Panel (b) compares the supplied reported level for all targets with graph distance among complete observed paths; these are not interchangeable measures.

ultimate-parent bucket from conservative name aliases. Candidate matches are ranked lexicographically by whether they occupy the expected preceding level, whether the supplied parent country matches, whether the original names match exactly, and whether the UIN matches. The canonical build accepts no fuzzy match.

The result is one occurrence-level edge for each target. Level-0 targets link to an explicit Indian ultimate-parent node. Below level 0, 1,500 of 1,650 relations link to an observed entity at the expected prior level, 72 link to an observed entity at another reported level, and 78 terminate at one of 18 normalized named-but-unscrapped parent nodes. These unresolved nodes retain their supplied country. Discarding the country would understate upstream exposure for some groups; treating the node as observed would overstate path completeness.

Occurrence edges are collapsed to a logical entity graph for subtree and gateway calculations. The parent-specific normalized graphs are directed and acyclic under the accepted links. The graph preserves alternative reported parents where they exist rather than forcing a tree.

3.2 Paths and depth

For every target, recursion follows observed parents toward the ultimate parent. A path is complete only when it reaches that root through observed entities. This succeeds for 1,687/1,834 targets (92.0%). A path that encounters a named missing node remains useful for country exposure up to that node but is not declared complete.

The supplied reported level and reconstructed graph distance are stored separately. Reported level may encode a hierarchy with nodes missing from the target table. Conversely, cross-channel name matching can create a shorter observed route than the reported sequence. Without external ownership charts, neither measure is relabeled as “true” depth.

3.3 Network concentration

Four concentration quantities answer different questions.

- The *modal-UIN share* is the fraction of a parent’s targets assigned to its largest UIN channel.
- The *largest observed subtree* counts normalized entity descendants below the observed entity with the largest descendant set, divided by the other entities in that parent network.
- The *named-node-inclusive subtree* also allows an unresolved named parent to be the top node, while still not calling the path complete.
- The *strict DAG dominator* removes an observed node and counts the entity nodes no longer reachable from the ultimate parent. It therefore respects alternative routes that bypass the candidate node.

The subtree summaries use the 24 parent groups with at least fifteen targets. That threshold prevents very small groups from determining a median while retaining most of the structural sample. Subtree language describes the mapped topology. It does not establish legal control, cash-flow dependence, or an operational chokepoint.

3.4 Geographic roles

The analysis distinguishes three jurisdiction roles. *Residence* counts normalized entities located in a jurisdiction. An *observed intermediary* is a unique observed entity that appears as the parent of at least one logical edge. A jurisdiction is *strictly upstream* of a target if it appears above the terminal target in the reconstructed path; the target’s own residence is excluded from its exposure set. India is excluded as the ultimate root in foreign-jurisdiction role and cover calculations.

A *gateway* is an observed level-0 entity. Gateway amplification is the number of normalized entity descendants below it. Jurisdiction means weight gateways; parent-equal means first average gateways within parent and then average parents. A robustness subset retains gateways for which no single direct-child branch contains a majority of descendants.

An edge is cross-border when the standardized child and immediate-parent country labels differ. The cross-border statistic excludes level-0 links to India, which would otherwise be cross-border by construction.

3.5 Weighting and robustness

For a binary target outcome y_i and parent $p(i)$, the pooled estimand is $N^{-1} \sum_i y_i$. The equal-parent estimand is

$$\frac{1}{P} \sum_{p=1}^P \left(\frac{1}{N_p} \sum_{i:p(i)=p} y_i \right).$$

The former describes target-weighted exposure in this file; the latter gives each parent bucket the same weight. Leave-one-parent-out estimates repeat the relevant pooled or fixed-effect calculation after excluding each parent in turn. Where units differ—entities, edges, paths, gateways, or parents—the paper labels the denominator explicitly rather than comparing percentages as though they shared an estimand.

Table 2: Illustrative parent architectures

Parent bucket	Raw rows	Entities	Countries	Reported max	Graph max	Modal UIN (%)	Complete paths (%)
Motherson	332	309	45	12	8.0	44.7	100
Wipro	504	212	51	8	7.0	50.2	88.7
Reliance Industries	277	196	22	2	2.0	34.7	100
Hindalco	81	81	25	12	6.0	95.1	46.9
Tata Communications	550	49	32	2	2.0	68.0	100
ONGC Videsh	41	41	18	2	2.0	29.3	100

Notes: Graph maximum is reconstructed distance among complete paths. Modal UIN is the largest UIN’s share of target occurrences. Complete paths are reported as a percentage of the parent’s targets. Rows are selected to illustrate contrasting architectures, not to rank firms.

4 Denominators and parent architecture

4.1 Raw-row weighting changes the sample composition

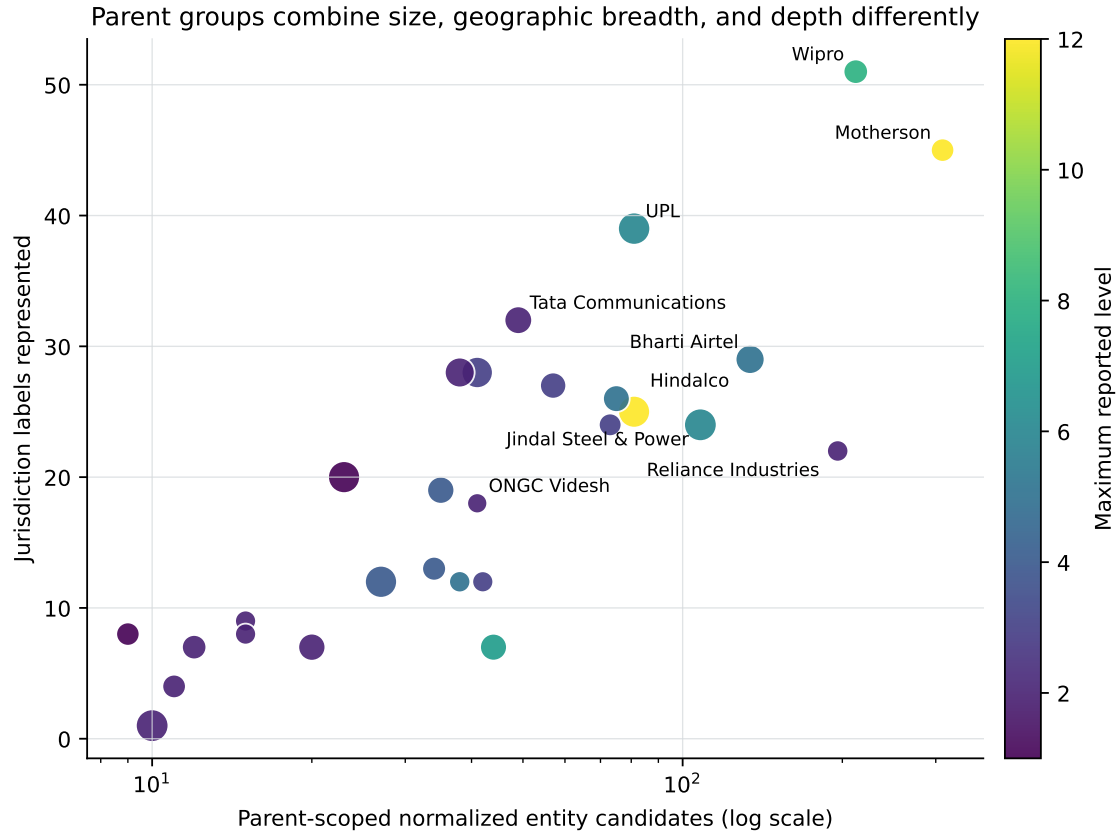
The denominator correction is empirically first order. Tata Communications contributes 550 source rows but only 49 parent-scoped normalized entities. Motherson contributes 332 rows but 309 normalized entities. Thus the former is largest by source rows while the latter is largest by network entities. Any statistic computed directly on source rows would overweight targets with longer or more duplicated financial-source histories.

The closeness of the target and normalized-entity totals is reassuring but does not make the units interchangeable. Four parent-scoped alias collapses are enough to matter for exact graph topology, and cross-parent shared names have a different interpretation from within-parent duplicates. The canonical tables therefore retain target and entity counts side by side.

4.2 Architecture is multidimensional

Figure 2 shows that group size, geographic breadth, reported depth, and channel concentration do not move together. Table 2 gives illustrative contrasts. Motherson is the largest entity network in the sample and has the greatest reported maximum depth, while Wipro spans more jurisdiction labels. Reliance Industries is large but shallow in reported levels. Tata Communications is much smaller as an entity network than its source-row count suggests. ONGC Videsh has many UIN channels relative to its target count and a shallow observed graph.

These comparisons do not support a one-dimensional complexity index. A group can be broad but shallow, deep but concentrated behind one entry channel, or large because one subtree contains many acquired or portfolio entities. Calling any one of those structures “more complex” would impose an unvalidated weighting across substantively different axes.



Marker area increases with the within-parent modal-UIN share.

Figure 2: Parent groups combine size, geographic breadth, and depth differently

Notes: Each marker is one supplied ultimate-parent bucket. Marker color is maximum reported level; marker area increases with the within-parent modal-UIN target share. Entity count uses the parent-scope normalized name-country key.

5 Dominant channels and large subtrees

The largest UIN within each parent accounts for 1,095/1,834 targets, or 59.7% pooled. Giving every parent equal weight produces 62.1%; the median parent is 61.2%, and the pooled leave-one-parent-out interval is narrow (Figure 8). Channel concentration is therefore not only an artifact of the largest parent bucket. At the same time, UIN is an administrative or mapping unit in this analysis, not a verified company or control node.

Graph concentration yields a complementary result. Across the 24 larger groups, the median largest observed subtree contains 43.2% of the other normalized entities. The median changes only modestly to 41.7% under the strict DAG dominator definition, which allows alternative paths to bypass the removed node. When named-but-unscraped parent nodes may sit at the top of a subtree, the median rises to 50.8%.

Figure 3 makes the conceptual separation visible. UIN and subtree shares are often close, but not always. A mapping channel can contain many targets whose observed graph is split, while a linked subtree can span more than one UIN. Missing parents can also reverse the apparent concentration of a particular group. The defensible result is that dominant mapped channels and large observed subtrees are common in this sample. It is not that every network has one legally dispositive company.

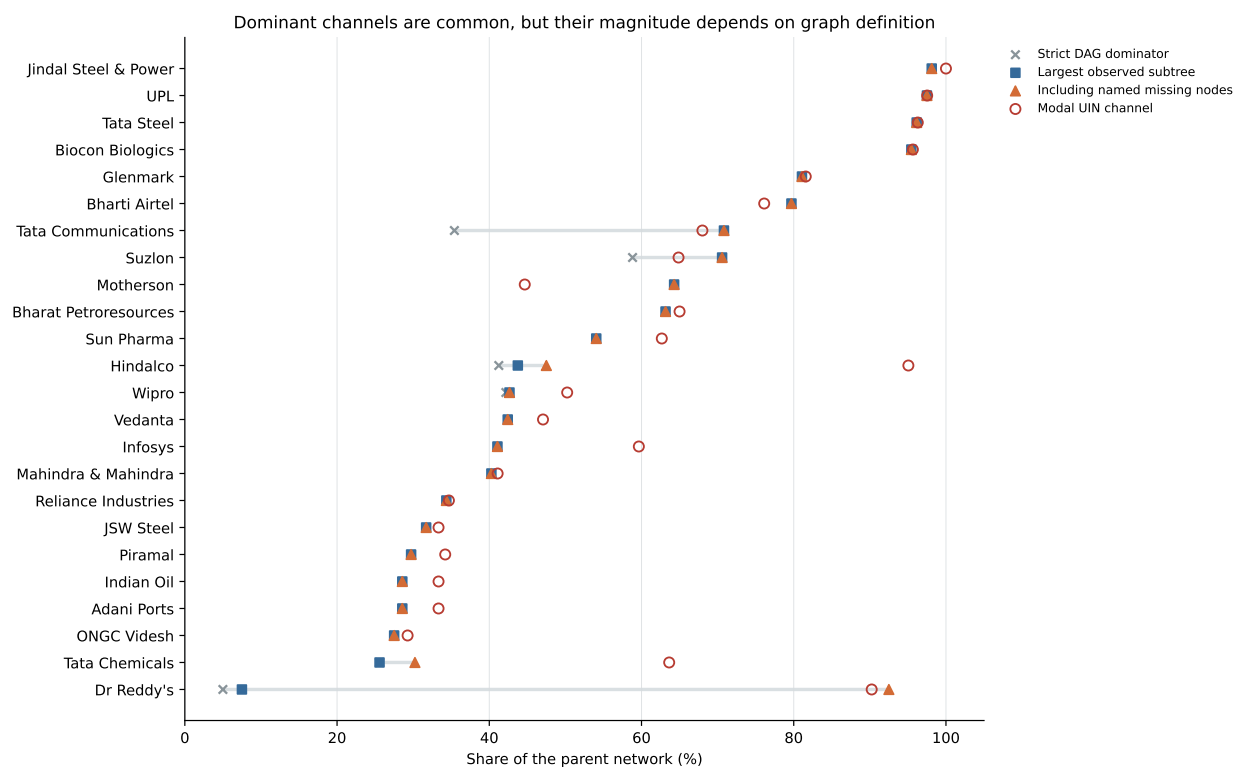


Figure 3: Channel concentration under alternative graph definitions

Notes: The sample is parent groups with at least fifteen targets. Subtree shares use normalized entity candidates and exclude the top node from the denominator. The strict DAG measure counts entities detached from the ultimate root after removing the candidate node. The hollow marker is the target-weighted modal-UIN share and therefore uses a different unit.

Table 3: Downstream amplification by selected first-hop jurisdiction

Gateway jurisdiction	Gateways	Parents	Mean	Median	Parent-equal mean	LOO mean range
Netherlands	13.0	12.0	34.8	12.0	37.1	21.2 to 39.7
Mauritius	17.0	11.0	17.9	5.0	19.7	12.4 to 19.5
United States	28.0	12.0	9.9	1.5	9.2	7.2 to 11.3
Singapore	18.0	12.0	7.5	3.5	7.4	6.0 to 8.5
United Arab Emirates	9.0	6.0	2.6	1.0	2.1	1.6 to 2.9

Notes: One gateway is one observed level-0 normalized entity. Mean and median weight gateways. The parent-equal mean first averages gateways within parent. The LOO range repeats the gateway-weighted mean after omitting each parent. The table shows selected jurisdictions; the full generated table contains every jurisdiction with at least three gateways.

6 Geography and intermediary structure

6.1 Cross-border ownership relations

Among 1,650 reported child–parent relations below level 0, 951 cross a standardized jurisdiction boundary. The pooled share is 57.6%, the equal-parent mean is 58.7%, and the leave-one-parent-out range is 56.1–61.8%. Recomputing with the raw supplied country labels gives essentially the same result (Table 6). This is one of the most definition-stable structural facts in the study.

The interpretation remains modest. A cross-border edge says only that the reported child and immediate parent carry different country labels. It does not reveal operating location, management, tax residence, legal control, or the purpose of the relation.

6.2 First-hop gateway amplification

Table 3 and Figure 4 treat each observed level-0 entity as a gateway and count normalized descendants below it. Dutch gateways have a mean of 34.8 descendants and median of 12.0. The parent-equal mean is 37.1, and the pooled leave-one-parent-out means range from 21.2 to 39.7. Restricting to gateways without a single majority child branch gives a mean of 37.8, so the average is not generated only by one immediate acquisition branch per gateway.

Mauritian gateways also show greater amplification than the United States comparison, although their median is well below their mean. United States gateways are numerous and span many parents, but typically have fewer descendants. The comparison illustrates why resident prevalence and network role must be measured separately.

6.3 Residence, intermediary prevalence, and path exposure

There are 403 unique observed parent nodes in the logical entity graph. Table 4 contrasts their distribution with resident entity counts and target paths. The United States is the largest resident location among the displayed jurisdictions. The Netherlands and Mauritius, however, have substantially more target paths below them relative to their resident counts. Figure 5 generalizes that comparison: points above the diagonal have more descendant paths than resident entities, while points below it are more terminal in this mapped topology.

The Netherlands is strictly upstream of 29.4% of target paths, compared with 20.9% after equal-parent weighting. The pooled value is therefore partly a function of the sizes of the parent networks using Dutch routes, but the role does not disappear when parents are balanced.

The greedy union of the Netherlands, United States, and Mauritius is strictly upstream of 1,140/1,834 target paths (62.2%). The equal-parent union is only 46.7%, and removing a large set

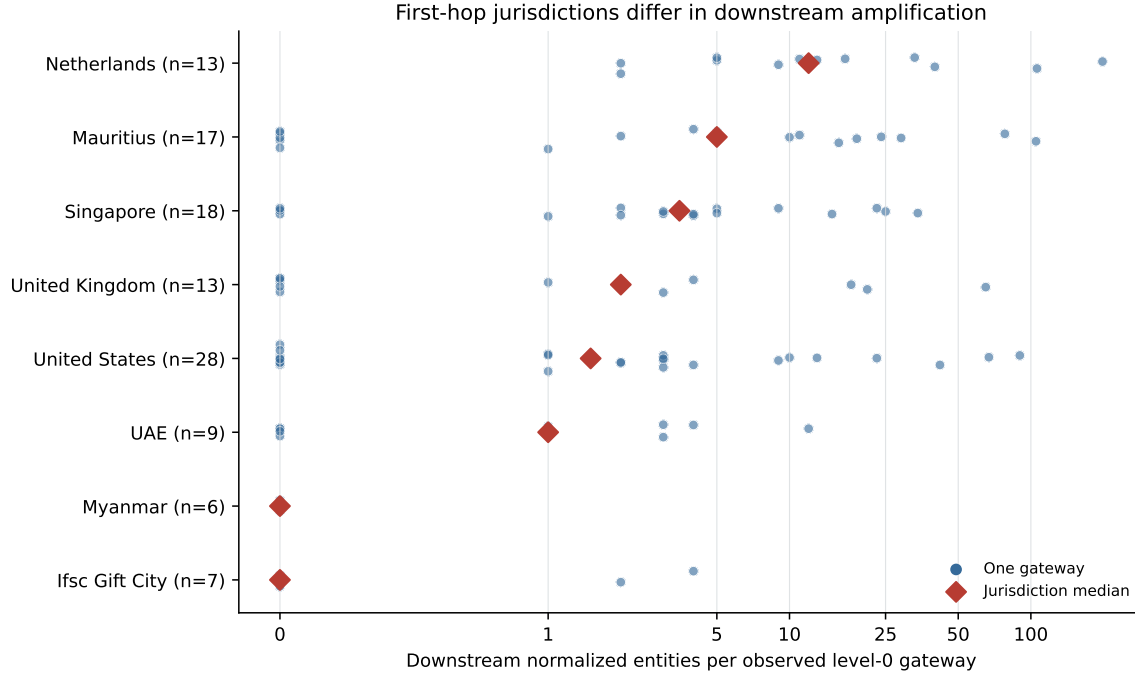


Figure 4: First-hop jurisdictions differ in downstream amplification

Notes: Each blue point is an observed level-0 gateway; diamonds mark jurisdiction medians. The horizontal axis uses a symmetric-log display to retain zero-descendant gateways. The plotted jurisdictions each have at least five observed gateways.

of venture-fund children gives 59.8%. The gap is substantive: the pooled cover is a descendant-weighted exposure statistic, not the share of unique intermediaries and not a counterfactual measure of what would be disconnected if three jurisdictions vanished.

6.4 Why a fixed “centre” category is not a core result

A maintained list of frequently discussed financial or intermediary locations is upstream of 61.8% of target paths, and the equal-parent estimate is 57.2%. But only 38.0% of unique observed intermediary nodes fall in the same list. The path figure repeats a shared upstream node for every descendant, and the list itself is imposed rather than observed. It also combines locations that may host operating, financing, acquisition, or holding entities. The canonical study therefore reports

Table 4: Selected jurisdiction roles under distinct denominators

Jurisdiction	Residents	Intermediaries	Paths below	Pooled (%)	Parent-equal (%)	Parents
United States	363	69.0	381	20.8	14.9	16.0
Netherlands	110	59.0	540	29.4	20.9	16.0
Mauritius	57.0	25.0	308	16.8	14.7	10.0
United Kingdom	99.0	39.0	219	11.9	10.2	16.0
Singapore	60.0	21.0	147	8.0	13.6	13.0

Notes: Residents are parent-scoped normalized entities. Intermediaries are unique observed parent nodes. Paths below count target paths on which the jurisdiction appears strictly upstream, including a supplied country on a named unresolved node. Pooled and parent-equal percentages use target paths; “Parents” is the number of parent buckets affected.

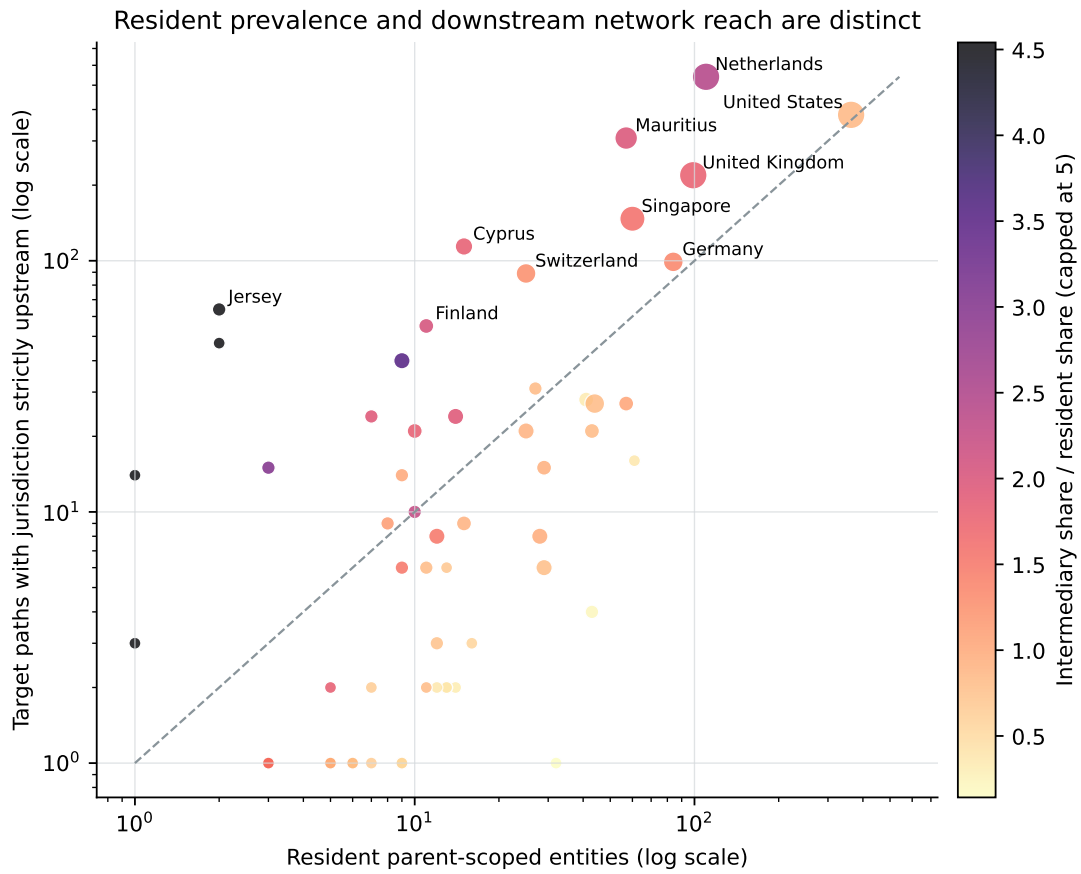


Figure 5: Resident prevalence and downstream network reach are distinct
Notes: Both axes are logarithmic. Marker area increases with the number of parent groups affected. Color is the jurisdiction's share of unique observed intermediary nodes divided by its share of resident entities, capped for display. India is excluded as the ultimate root.

specific measured roles and keeps the aggregate classification as a labeled sensitivity analysis.

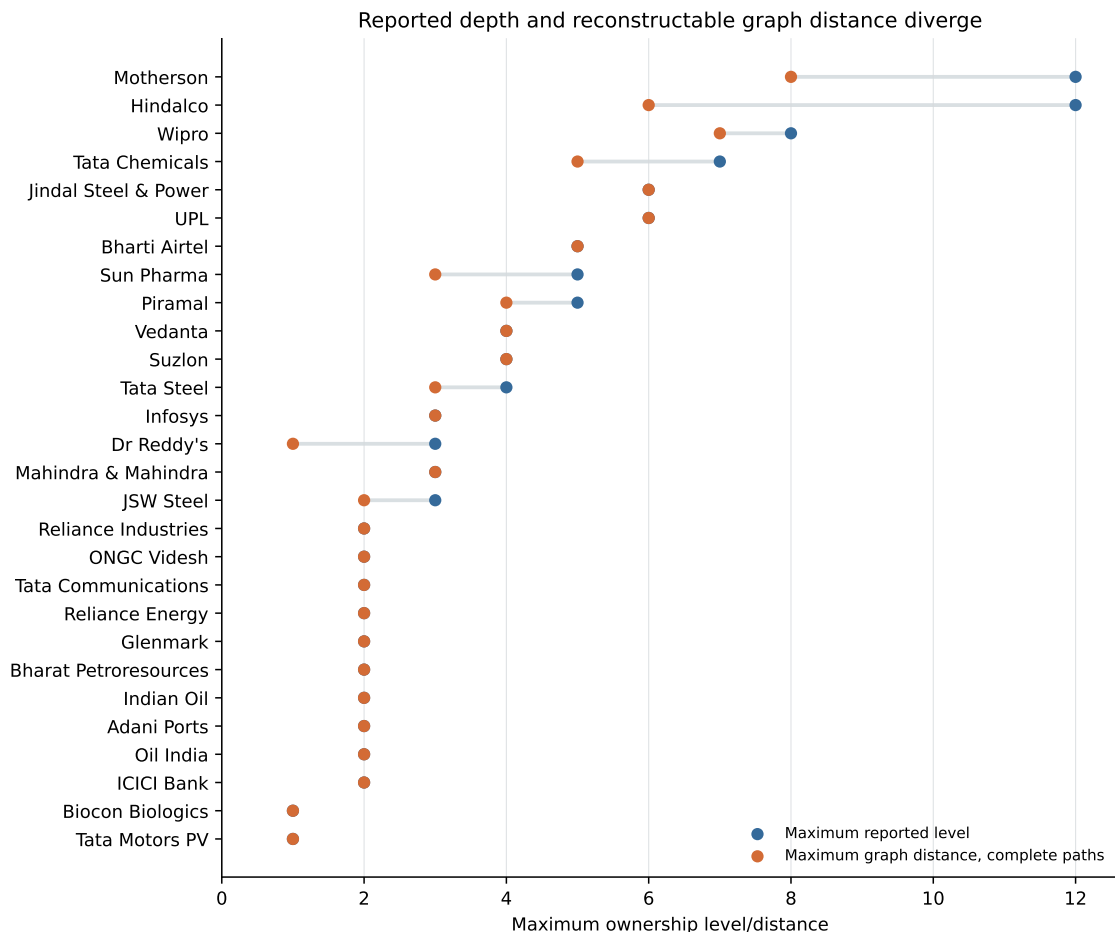


Figure 6: Reported depth and reconstructable graph distance diverge

Notes: Each row is one parent bucket. Blue markers show maximum supplied reported level. Orange markers show maximum reconstructed distance among paths that reach the ultimate parent through observed entity nodes. The connecting segment is descriptive; neither endpoint is designated the legal truth.

7 Reported depth and observed graph distance

Nearly half of targets are reported below the first foreign layer: 871/1,834 (47.5%) have reported level 2 or higher. The apparently deep tail is less stable. Reported level 5 or higher contains 342 targets (18.6%), while the equal-parent mean is only 7.3%. Among complete paths, just 7.4% have reconstructed distance 5 or greater.

The measures disagree directly for 236/1,687 complete paths (14.0%). Motherson contributes 88.1% of these mismatches. Figure 6 shows that the maximum reported and reconstructed depths can diverge sharply within a group. Reported level may preserve omitted layers; reconstructed distance may be a lower bound. The concentration of disagreements in one parent also means that a pooled depth correction cannot be assumed to apply uniformly.

The sector result provides a falsification exercise. Of 508 targets labeled general manufacturing, 242 are reported at level 5 or deeper. That looks like a broad sector-depth association until parent composition is examined: Motherson and Hindalco account for 240/242 (99.2%) of the deep manufacturing targets. The paper therefore rejects “manufacturing subsidiaries are generally deeper” as a sample-wide conclusion.

Table 5: Financial coverage and quality diagnostics

Gate or diagnostic	Numerator	Denominator	Value	Unit
Targets with any source	847	1,834	46.2	targets
Targets with parsed balance sheet	444	1,834	24.2	targets
Targets ready at least once	265	1,834	14.4	targets
Ready rows	560	3,567	15.7	preferred target-years
Sign-plausible ready rows	521	560	93.0	ready rows
P&L-valid rows	105	560	18.8	ready rows
Ready rows with blank units	516	560	92.1	ready rows
Demonstrated same-URL/different-name reuse clusters	33.0	51	64.7	repeated-signature clusters
Ready rows unflagged by broad duplicate/sign screen	460	560	82.1	not a validated final sample

Notes: The value column is a percentage except for diagnostics whose unit states otherwise. Repeated numeric signatures are fiscal-year/currency/ units/assets/liabilities/equity clusters. Demonstrated reuse additionally requires the same URL, different target names, and byte-identical core evidence text.

8 Ownership fields, financial coverage, and structural proxies

8.1 Recorded stakes and the ownership-depth gradient

Stake information cannot support a complete control-weighted network. Every observed level-0 target has missing mapping stake, and 406 of 1,650 nonroot relations record exactly zero (24.6%). Where a preferred financial row contains both mapping stake and an AOC-1 shareholding measure, each comparable zero-stake case is contradicted by a positive AOC value (Appendix C). The zeros are therefore treated as unknown, not as literal absence of ownership.

Restricting to positive-stake edges produces a superficially strong gradient: the full-ownership rate is 56.4% at reported level 1 and 82.8% at level 2 or deeper, a pooled difference of 26.4 percentage points. The equal-parent difference is only 2.3 points, and excluding Reliance reduces the pooled gap to 6.1 points. The stable conclusion is a parent and portfolio composition effect, not that ownership mechanically strengthens with depth. Cumulative ownership along full paths is excluded because it would condition on positive recorded stakes and treat unresolved zeros selectively.

8.2 Financial attrition and extraction reuse

Figure 7 and Table 5 document the financial coverage funnel. A source is located for 847/1,834 targets, a balance sheet is parsed for 444, and only 265 targets are ready at least once. The preferred panel contains 560 ready target-years; 521 pass a positive-assets and nonnegative-liabilities gate, and only 105 have a testable valid profit-and-loss identity. Units are blank for 516/560 ready rows.

Duplicate-evidence checks reinforce the limitation. There are 51 repeated signatures defined jointly by fiscal year, currency, units, assets, liabilities, and equity; they cover 104 parsed rows. In 33 clusters, the same source URL is attached to different entity names and the core evidence text is byte-identical. A broad union of repeated-signature and sign flags leaves 460 ready rows, but that remainder is a diagnostic, not a validated analysis sample.

Readiness itself varies greatly across parents, and 5 parents have no ready target. The raw association between reported depth 3+ and readiness is -6.6 percentage points; a parent fixed-effect coefficient is -2.4 points, with leave-one-parent-out estimates from -6.6 to -0.8 points. This is attenuation rather than elimination, but selection, unit missingness, and extraction reuse make a financial-performance interpretation indefensible.

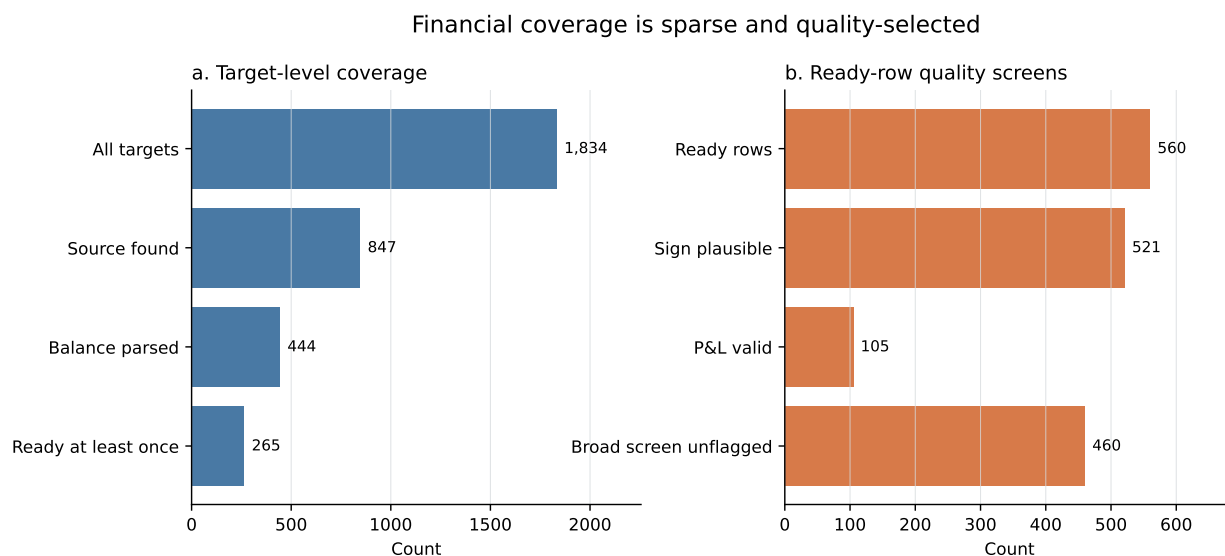


Figure 7: Financial coverage is sparse and quality-selected

Notes: Panel (a) counts structural targets at successive coverage gates. Panel (b) counts preferred target-year rows. The broad screen flags either a repeated numeric signature or an implausible sign combination; its unflagged remainder has not been manually validated.

8.3 A limited name-based structural proxy

One supporting result survives a within-cell test. 177 targets contain “Holding(s),” “Holdco,” or “Investment(s)” in the normalized name. 64.4% of flagged targets hold at least one observed child, compared with 17.6% of other targets. Residualizing both the outcome and name flag within parent-by-country cells gives a 36.3-point association. This suggests that names can be a useful high-precision, low-recall structural proxy when edges are absent. It does not verify a legal role, and external role labels and held-out-parent validation would be required before using it as a general classifier.

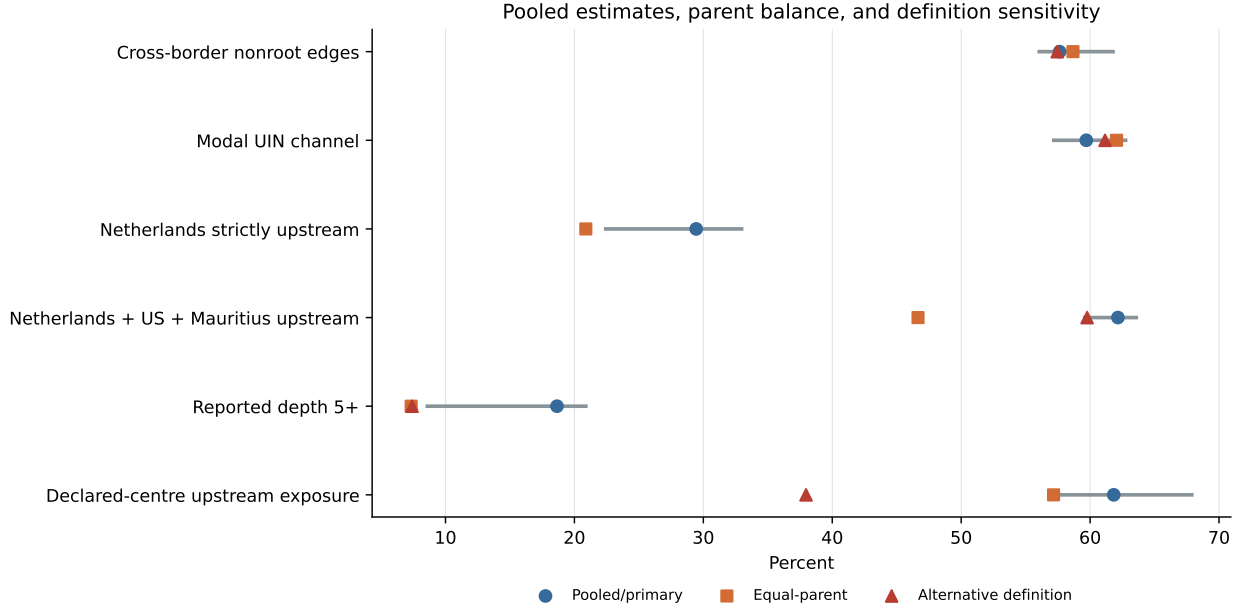


Figure 8: Pooled estimates, parent balance, and definition sensitivity

Notes: Grey segments are leave-one-parent-out ranges where applicable. Blue circles show pooled or primary estimates, orange squares show equal-parent estimates, and red triangles show the labeled alternative definition. Rows share a percent scale but not necessarily the same observation unit; Table 6 states each unit.

9 Robustness, heterogeneity, and evidence classification

Figure 8 and Table 6 collect the main robustness results on one scale. Three patterns are notably stable. The cross-border edge share barely moves under equal-parent weighting, raw country labels, or leave-one-parent-out checks. Modal-UIN concentration also remains high under parent balance. Dutch gateway amplification remains large under equal-parent weighting, omission of any one parent, and the no-majority-branch restriction.

Other estimates change meaningfully. The three-country upstream union and reported deep share fall under equal-parent weighting. The full-ownership depth gap shrinks sharply under parent correction. Financial readiness retains a small negative within-parent depth association, but its LOO range and severe measurement attrition preclude substantive inference. The maintained-centre exposure is much smaller when the unit is unique observed intermediary nodes rather than descendant paths.

No formal probability sample underlies the 28 parent buckets, so the paper does not present sampling-error p -values as though they supported population inference. Robustness here means sensitivity to measurement and composition: unique entities versus repeated targets, pooled versus parent-balanced weights, omission of any one parent, alternative graph definitions, fund exclusions, and explicit missingness rules.

Table 6: Canonical robustness matrix

Estimand	Unit	Primary	Parent-equal/FE	LOO range	Alternative	Definition
Cross-border nonroot edges	%	57.6	58.7	56.1 to 61.8	57.5	Raw-label sensitivity
Modal UIN channel	% of targets	59.7	62.1	57.2 to 62.8	61.2	Median parent
Largest observed subtree	median parent %	43.2	–	–	41.7	Strict DAG dominator
Dutch gateway amplification	mean descendants	34.8	37.1	21.2 to 39.7	37.8	No majority child branch
Netherlands strictly upstream	% of targets	29.4	20.9	22.4 to 33.0	–	Descendant/path estimand
Netherlands + US + Mauritius upstream	% of targets	62.2	46.7	59.6 to 63.6	59.8	Exclude venture-fund children
Reported depth 5+	% of targets	18.6	7.3	8.6 to 20.9	7.4	Reconstructed depth 5+, complete paths
Full-ownership depth gap	percentage points	26.4	2.3	–	6.1	Exclude Reliance
Financial-ready deep gap	percentage points	-6.6	-2.4	-6.6 to -0.8	–	Parent FE in equal-parent column

Notes: “Primary” is pooled unless the row labels a median-parent estimand. FE denotes a fixed-effect coefficient. LOO repeats the relevant pooled or fixed-effect estimator after dropping each parent. Alternatives are not always percentages of the same unit and should be read with the definition column.

10 Limitations and interpretation boundary

The study’s limitations are part of its empirical result rather than a generic disclaimer.

- **Sample scope.** The parent buckets are supplied, not sampled from a documented universe. Results describe these groups and cannot establish prevalence across Indian multinational firms.
- **Entity identity.** Normalized names and countries are candidate identities. They cannot distinguish every alias, reorganization, joint exposure, or legally separate same-name company without registration identifiers.
- **Hierarchy completeness.** Named missing parents truncate paths, and reported levels disagree with observed graph distance. External ownership charts are required to adjudicate which representation is legally or historically correct.
- **UIN semantics.** The data establish mapping-channel concentration but do not validate the meaning of UIN substrings or their use as dates, office codes, or ownership types. Vintage claims are therefore excluded.
- **Jurisdiction labels.** A country label is not proof of operating location, management, tax residence, or entity function. The data cannot identify a tax, treaty, financing, acquisition, governance, or operating motive.
- **Ownership measurement.** Missing level-0 stakes and contradicted zero values prevent control weighting and cumulative-ownership inference.
- **Financial selection.** Sparse coverage, blank units, mixed currencies, invalid signs, limited P&L identities, and demonstrated evidence reuse rule out population financial comparisons.
- **Time and current status.** The source combines reported structures and source years but does not validate a common current ownership date for every edge. The network is a reconstruction of the supplied report, not a claim about present legal control.

Accordingly, unusual topology is not evidence of wrongdoing. The paper does not label any company, jurisdiction, route, or transaction fraudulent, illegal, evasive, abusive, or improper. Legal, regulatory, institutional, and tax interpretations require external evidence and are outside the identified design.

11 Conclusion

The canonical reconstruction changes the object of study from a financial- source panel into an explicit set of target occurrences, normalized entity candidates, edges, and ownership paths. That correction is not cosmetic: it reverses parent rankings and prevents long source histories from masquerading as large corporate networks.

Within the supplied sample, parent architectures are genuinely heterogeneous. Dominant mapping channels and large subtrees are common, but their magnitudes depend on whether the unit is a UIN, observed subtree, named-node-inclusive subtree, or strict graph dominator. The Netherlands and Mauritius occupy intermediary and gateway roles that are not reducible to resident counts, and the majority-cross-border-edge result is stable to parent reweighting. These are defensible facts about the mapped topology.

The negative findings are equally important. Deep manufacturing is almost entirely a two-parent pattern. The pooled ownership-depth gradient is largely composition. Reported depth is not observed graph distance. The financial layer cannot support comparative performance claims. A credible study of corporate ownership networks therefore depends as much on refusing the wrong denominator and interpretation as on finding a large statistic.

Future work can extend the evidence boundary only with new validated inputs: official entity identifiers, dated ownership charts, documented UIN semantics, verified stake fields, harmonized financial units, and a transparent sampling frame. Until then, the contribution is a reproducible, falsifiable map of the reported network structure in this dataset.

A Exact estimands and counting rules

A.1 Units

The pipeline distinguishes the following units throughout:

- a source candidate row for source and extraction audits;
- a preferred target-year for financial readiness and accounting checks;
- a target occurrence for reported path exposure;
- a parent-scoped normalized entity for within-group architecture;
- a global normalized entity for cross-parent unique-company sensitivity;
- a nonroot occurrence edge for direct child–parent geography;
- a logical entity edge for graph topology;
- a complete observed target path for reconstructed distance;
- an observed level-0 gateway for descendant amplification; and
- an ultimate-parent bucket for equal-parent and omission checks.

No table silently substitutes raw rows for entities or target paths for unique intermediaries. Percentages in generated tables always travel with a stated numerator, denominator, or unit.

A.2 Missing values and duplicates

Blank country or name fields are never imputed through fuzzy similarity. Named missing parents are kept as unresolved name–country nodes. Recorded zero stakes are treated as unknown for control analyses. Structural fields are required to be invariant within target identifier. Financial duplicates are screened using exact signatures over fiscal year, currency, units, assets, liabilities, and equity; demonstrated evidence reuse additionally requires a shared source URL, different entity names, and identical core evidence text.

A.3 Subtree and path denominators

For parent p with normalized entity set V_p and candidate observed node v , the observed-subtree share is

$$S_{pv} = \frac{|\text{Desc}(v) \cap V_p|}{|V_p| - 1}.$$

The reported parent statistic is the maximum S_{pv} over observed parent nodes. The named-node-inclusive variant expands the candidate set to unresolved named parents. The strict dominator variant removes v and counts only entity nodes that cease to be reachable from the ultimate root; alternate graph paths therefore reduce its value.

For an upstream-jurisdiction result, each terminal target contributes at most one indicator per jurisdiction, regardless of how often that jurisdiction appears on its path. A shared ancestor still appears once for every terminal target below it, which is why target-path exposure is reported separately from unique intermediary-node prevalence.

B Depth audit

Table 7: Reported depth, reconstructed distance, and parent concentration

Depth diagnostic	Numerator	Denominator	Value (%)	Unit
Reported level 2+	871	1,834	47.5	target occurrences
Reported level 5+	342	1,834	18.6	target occurrences
Reported level 5+, equal-parent	—	28	7.3	parent mean
Reconstructed distance 5+	125	1,687	7.4	complete paths
Reported/reconstructed mismatch	236	1,687	14.0	complete paths
Motherhood share of mismatches	208	236	88.1	mismatching complete paths
Manufacturing targets at reported level 5+	242	508	47.6	manufacturing target occurrences
Motherhood + Hindalco share of deep manufacturing	240	242	99.2	reported level-5+ manufacturing targets

Notes: Reported measures use all structural targets. Reconstructed distance and the mismatch rate use only complete observed paths. The equal-parent deep rate averages the parent-specific reported level-5+ shares.

C Ownership-field audit

Table 8: Stake coverage and ownership-depth diagnostics

Ownership diagnostic	Numerator	Denominator	Value	Unit
Level-0 stakes missing	184	184	100	level-0 targets
Nonroot recorded zero stakes	406	1,650	24.6	nonroot edges
Positive nonroot stakes	1,244	1,650	75.4	nonroot edges
Rows with mapping stake and AOC share	45.0	3,567	1.3	preferred target-years
Zero mapping stake but positive AOC share	7.0	7	100	comparable zero-stake rows
Stake/AOC difference above 1 pp	8.0	45	17.8	comparable rows
Full ownership at level 1	292	518	56.4	positive-stake edges
Full ownership at level 2+	601	726	82.8	positive-stake edges
Pooled level-2+ minus level-1 gap	—	1,244	26.4	percentage points
Equal-parent level-2+ minus level-1 gap	—	27	2.3	percentage points
Gap excluding Reliance	—	1,081	6.1	percentage points

Notes: Full ownership is defined as a recorded positive stake of at least 99.5 percent. Level-0 stakes are missing and nonroot zeros are not imputed. The equal-parent gap averages parent-specific rates; groups without both depth cells do not identify a paired contrast.

D Evidence disposition

Table 9: Canonical classification of discovered and reviewed findings

Class	Result	Disposition
Core	Denominator correction	Raw rows are source observations, not firms.
Core	Multidimensional parent architecture	Size, geography, depth, branching, and channels are distinct axes.
Core	Dominant channels and large observed subtrees	Report UIN, observed-subtree, named-node, and DAG estimands separately.
Core	Gateway-jurisdiction topology	Dutch and Mauritian roles survive unique-node, gateway, parent-weighted, and LOO checks.
Core	Depth measurement and parent concentration	Reported depth is not observed graph distance; pooled deep patterns are parent-driven.
Supporting	Cross-border nonroot edges	Stable under parent balance and leave-one-parent-out checks.
Supporting	Stake-zero and ownership-composition audit	Recorded zeros are unsafe; the pooled ownership-depth gradient is composition.
Supporting	Financial selection and extraction audit	Coverage and parsing quality preclude population financial conclusions.
Supporting	Holding-type name proxy	Large within parent-country association, pending external role validation.
Descriptive	Three-country descendant cover	Large path-weighted exposure but materially lower with equal-parent weighting.
Fragile	Fixed 19-jurisdiction centre exposure	Maintained list and descendant weighting determine the magnitude.
Fragile	Cumulative no-dilution claim	Positive-stake complete chains are selected and zeros are unresolved.
Unresolved	UIN substring semantics and vintage	Requires an authoritative RBI specification.
Rejected	General deep-manufacturing pattern	Two parents account for nearly all deep manufacturing targets.
Rejected	Financial health by depth or jurisdiction	Selection, units, duplicate extraction, and small valid samples preclude inference.
Unresolved	Tax, legal, regulatory, or misconduct mechanism	Topology alone does not identify purpose, legality, or causation.

Notes: “Core” requires agreement across the independent research streams plus robustness to the relevant denominator, parent-weighting, and hierarchy checks. Supporting results are defensible but not the paper’s main claim. Fragile, unresolved, and rejected claims are excluded from the core narrative.

E Reproducibility

From the repository root, the complete analysis and verification run with:

```
python -m src.analysis.run_pipeline --clean
python -m src.analysis.validate
pdflatex -interaction=nonstopmode -halt-on-error \
  -output-directory=paper paper/main.tex
pdflatex -interaction=nonstopmode -halt-on-error \
  -output-directory=paper paper/main.tex
```

The analysis command checks immutable input hashes and exact central invariants before writing any reported output. The validator rebuilds into a temporary directory and byte-compares deterministic tables, PNG figures, metrics, and generated research reports. Figure PDFs and the compiled paper may contain run-time PDF metadata; their content is checked by compilation and visual inspection rather than byte equality. Full definitions and the output map appear in `REPLICATION.md`.